

MOSTAFA ALI

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EDUCATION

Bachelor of Computing (Honours), Computer Science Sep 2024 - Apr 2028

University of Guelph — Guelph, ON

Coursework: Data Structures, Algorithms, Operating Systems, Databases, Statistics

WORK EXPERIENCE

BI/Data Analyst Intern — Strategy Team | Kensington May 2026 - Present

SQL | Power BI | Microsoft Fabric | Lakehouse | SSMS | PySpark | Azure DevOps | Python | SharePoint | Asana

- Built reusable Power BI semantic models by writing extensive SQL across SSMS and Microsoft Fabric Lakehouse — authoring views, transformation logic, and 6+ DAX measures — with source control and deployment managed through Azure DevOps
- Translated business questions from senior strategy stakeholders into stakeholder-facing Power BI dashboards by partnering directly in weekly working sessions, surfacing booking trends, customer segmentation, and revenue and market-expansion KPIs from large transactional datasets
- Accelerated routine query development and data-quality checks by building an AI-assisted workflow that connects Claude to the SQL environment through a Model Context Protocol (MCP) server, cutting manual iteration time on recurring analysis
- Diagnosed data pipeline failures by tracing missing or inconsistent records back to source transformation logic across SSMS and Microsoft Fabric Lakehouse, then built corrective ETL pipelines to resolve recurring data-quality gaps

PROJECTS

PantryChef AI — AMD Developer Hackathon 2026

Python | FastAPI | Llama 3.2 11B Vision | torchao int8 | SQLite | USDA FoodKeeper | React | Zustand | ROCm 7.0 | PyTorch 2.6

- Deployed Llama 3.2 11B Vision on a cloud-hosted AMD Instinct MI300X GPU instance (ROCm 7.0, PyTorch 2.6) with SDPA attention and torchao int8_weight_only quantization, halving VRAM footprint from 22 GB to 11 GB while maintaining inference quality
- Engineered a multi-stage data pipeline combining vision model output, a Python difflib fuzzy-matching layer, and USDA FoodKeeper shelf-life data with unit-aware conversion (Days/Weeks/Months) to auto-populate a live SQLite pantry inventory
- Designed a structured LLM inference system applying prompt engineering with Divide & Conquer prompting and max_new_tokens=1200 to reliably generate 5 distinct zero-waste recipes per session, prioritizing near-expiry ingredients from the live pantry state
- Architected a background model loading system using Python threading with a /health endpoint and threading.Lock for serialized GPU inference, enabling FastAPI to serve requests instantly while the 11B model loaded asynchronously

AI Book Reader — Multimodal PDF-to-Audiobook System 2026

Python | Streamlit | Google Gemini 1.5 Flash | Kokoro-82M | PyMuPDF | NLTK | PortAudio | threading

- Built a multimodal book-to-audiobook pipeline by integrating Google Gemini 1.5 Flash vision with structured prompt engineering (context injection + SKIP escape-hatch) to generate audio-natural descriptions for embedded PDF images, read inline alongside narrated text
- Delivered gapless real-time audio playback by deploying Kokoro-82M local TTS and architecting a Producer-Consumer pipeline (Python threading, queue.Queue, PortAudio at 48 kHz) that synthesizes the next sentence while the current sentence plays
- Validated Gemini 1.5 Flash Vision API integration through systematic testing of response accuracy and error handling across edge cases (corrupted pages, missing images, malformed PDFs)

Risk & Security Control Management System (RSCMS) 2025

Python | Flask | PostgreSQL | Power BI | LangChain | ChromaDB | Claude API | psycopg2 | Docker

- Modeled 5,000+ synthetic security incidents by designing a 5-table PostgreSQL schema with composite unique constraints, computing MTTR/MTTD via EXTRACT(EPOCH FROM ...) in SQL and surfacing KPIs across 6 DAX measures and 9 Power BI visuals
- Exposed live analytics to a Power BI dashboard by building a 9-endpoint Flask REST API on psycopg2 with query-param filtering, then layered a LangChain RAG pipeline (ChromaDB + Claude API) with prompt engineering for natural-language querying of NIST 800-53 and ISO 27001 controls
- Wrote and ran unit and integration tests against all 9 Flask REST API endpoints to debug query-param filtering issues and verify response correctness across edge cases
- Produced statistically meaningful MTTR/MTTD averages for executive reporting by generating 5,000+ synthetic incidents with Faker, using realistic detection-lag and response-time distributions across a 90-day window

TECHNICAL SKILLS

GenAI & Agentic AI: LangChain, Claude API, Gemini API, Google ADK, RAG pipelines, prompt engineering, ChromaDB

ML & Data Science: Scikit-learn, PyTorch, Keras, TensorFlow, XGBoost, Pandas, NumPy, SciPy, NLTK

Languages: Python, C, SQL, JavaScript, R, Bash

Data & BI: PostgreSQL, SSMS, SQLite, PySpark, Power BI, Microsoft Fabric, Lakehouse, DAX, ETL, IBM Cloud, Azure, AWS

Dev & Tools: Docker, FastAPI, Flask, Streamlit, Azure DevOps, GitLab CI/CD, Git, Linux, SharePoint, Asana, Cursor AI, GitHub Copilot

CERTIFICATIONS

IBM Deep Learning w/ PyTorch, Keras & TF | IBM / Coursera 2026 AWS Cloud Practitioner | Amazon Web Services 2026

Building with the Claude API | Anthropic Apr 2026 AI Fluency Framework & Foundations | Anthropic Apr 2026

Intro to Splunk | Splunk Apr 2026 Microsoft Security Essentials | Microsoft Jan 2026

GDPR Foundations | Measured Collective Feb 2026